$$\int_{3\pi/4}^{\pi} \tan\theta \, d\theta$$

$$\int_{\theta=3\pi/4}^{\theta=\pi} \frac{\sin\theta}{\cos\theta} \, d\theta$$

$$u = \cos\theta$$

$$du = -\sin\theta \, d\theta$$

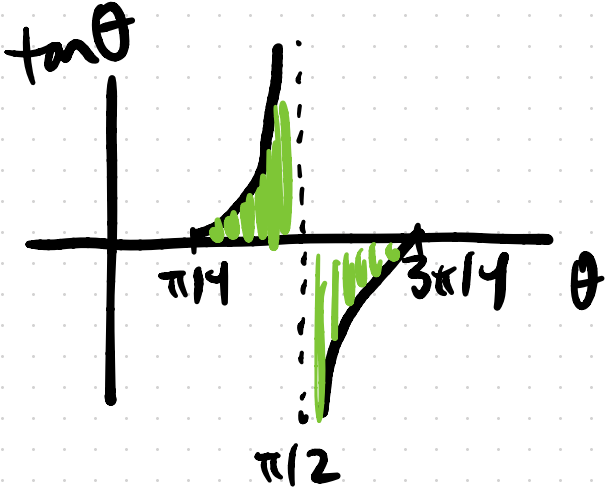
$$-du = \sin\theta \, d\theta$$

$$\int_{-\sqrt{2}/2}^{-1} \left(-\frac{1}{u}\right) du$$

$$-\ln|u| \Big|_{-\sqrt{2}/2}^{-1} = 0 - \left(-\ln\frac{\sqrt{2}}{2}\right) = \ln\frac{\sqrt{2}}{2}$$

$$\int_{\pi/4}^{3\pi/4} \tan\theta \, d\theta$$

$$-\ln|u| \Big|_{\sqrt{2}/2}^{-\sqrt{2}/2} = \cancel{0}$$



$\int_{\pi/4}^{3\pi/4} \tan\theta \, d\theta$ is not defined because $\tan\theta$ has a vertical asymptote at $\theta = \pi/2$

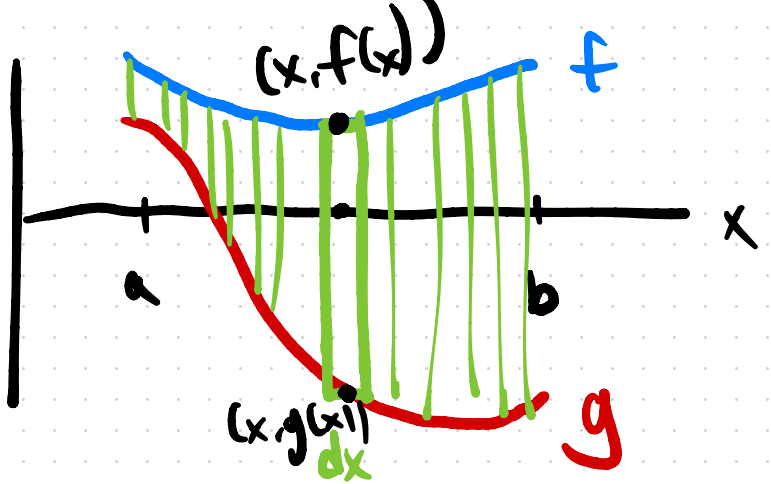
Areas between curves (6.1)

This chapter concerns applications of integration. The typical setup is as follows.

- We have some quantity (area, volume, etc.) we want to compute.
- We write this quantity as a definite integral.
- We use antidifferentiation to compute this definite integral (if possible).



Suppose we have two functions f, g defined on the interval $[a,b]$, and $f(x) \geq g(x)$ for all x on this interval. What is the area between the their graphs on this interval?



"Subdivide" the interval into infinitely thin rectangles of "width" dx, and height $f(x) - g(x)$

"Sum" up the areas of the rectangles

$$\int_a^b (f(x) - g(x))\,dx$$

More rigorously, the area is

$$\lim_{n\to\infty} \sum_{i=1}^{n} (f(x_i^*) - g(x_i^*))\Delta x$$

$$= \int_a^b (f(x) - g(x))\,dx$$

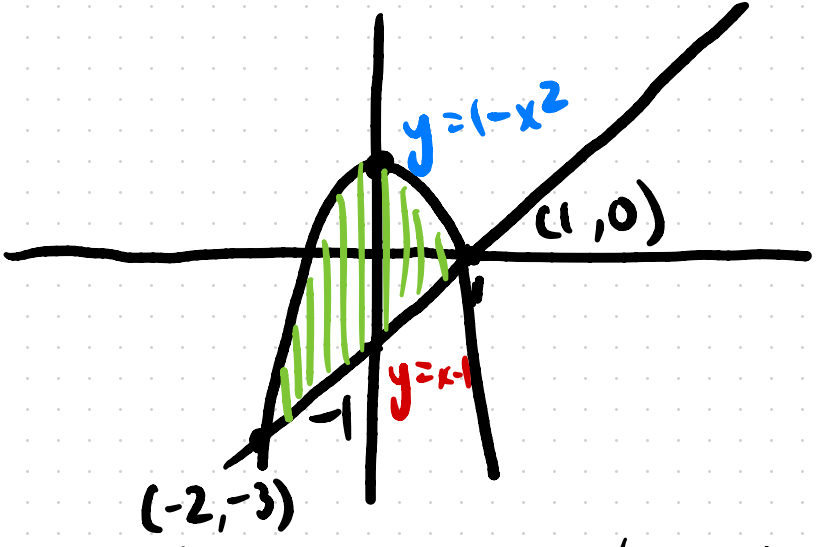
by definition

Theorem: If $f(x) \geq g(x)$ on for all $a \leq x \leq b$, then the area between the graphs of f and g on this interval is

$$\int_a^b (f(x) - g(x))\, dx$$

What is the area enclosed by the graphs of $y = x - 1$ and $y = 1 - x^2$?

$y = a(x-h)^2 + k$



Need to find the points of intersection.

Solve the system of equations

$y = x - 1$

$y = 1 - x^2$

$x - 1 = 1 - x^2$

$x^2 + x - 2 = 0$

$(x+2)(x-1) = 0$

$x = -2, 1$

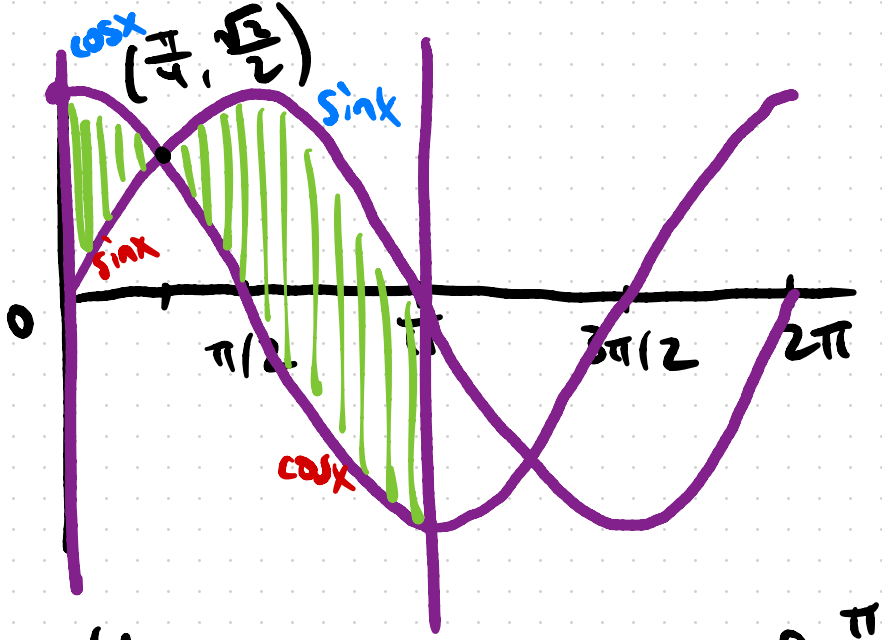
Area is

$$\int_{-2}^{1} \left((1 - x^2) - (x - 1) \right) dx = \int_{-2}^{1} \left(-x^2 - x + 2 \right) dx$$

$$= -\frac{1}{3}x^3 - \frac{1}{2}x^2 + 2x \Big|_{-2}^{1}$$

$$\left(-\frac{1}{3} - \frac{1}{2} + 2\right) - \left(\frac{8}{3} - 2 - 4\right) = \frac{9}{2}$$

What is the total area of the regions bounded by $y = \sin x$, $y = \cos x$, $x = 0$, $x = \pi$?



$\sin x = \cos x \quad x \in [0, \pi]$

$x = \pi/4$

$$\int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi} (\sin x - \cos x)\,dx$$

$$\sin x + \cos x \Big|_0^{\pi/4} + \left(-\cos x - \sin x\right) \Big|_{\pi/4}^{\pi}$$

$$\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) - (1) + \left(1\right) - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right)$$

$$2\sqrt{2}$$

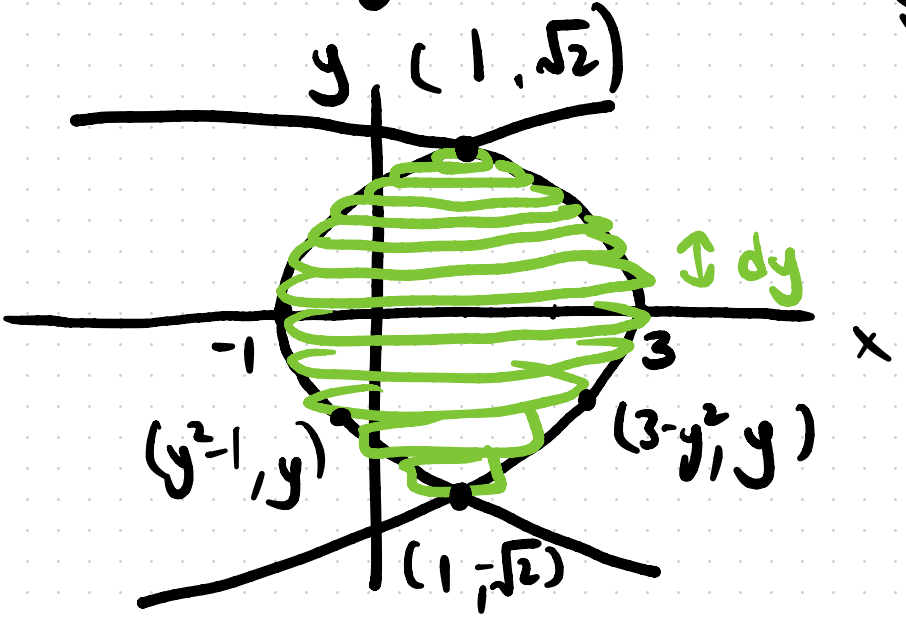
Sometimes, it is easier to integrate with respect to y.

Theorem: If $f(y) \geq g(y)$ on for all $a \leq y \leq b$, then the area between the graphs of f and g on this interval is

$$\int_a^b (f(y) - g(y))\, dy$$

What is the area enclosed by the graphs of $x = y^2 - 1$ and $x = 3 - y^2$?

$$y^2 - 1 = 3 - y^2$$
$$2y^2 = 4$$
$$y = \pm\sqrt{2}$$



"Area" of a horizontal rectangle $= \left((3 - y^2) - (y^2 - 1)\right) dy$

$$\int_{-\sqrt{2}}^{\sqrt{2}} \left((3-y^2)-(y^2-1)\right)dy = \int_{-\sqrt{2}}^{\sqrt{2}} (-2y^2+4)\,dy$$

$$= -\frac{2}{3}y^3 + 4y \Big|_{-\sqrt{2}}^{\sqrt{2}}$$

$$\left(-\frac{4\sqrt{2}}{3}+4\sqrt{2}\right) - \left(\frac{4\sqrt{2}}{3} - 4\sqrt{2}\right)$$

$$-\frac{8\sqrt{2}}{3} + 8\sqrt{2} = \frac{16\sqrt{2}}{3}$$